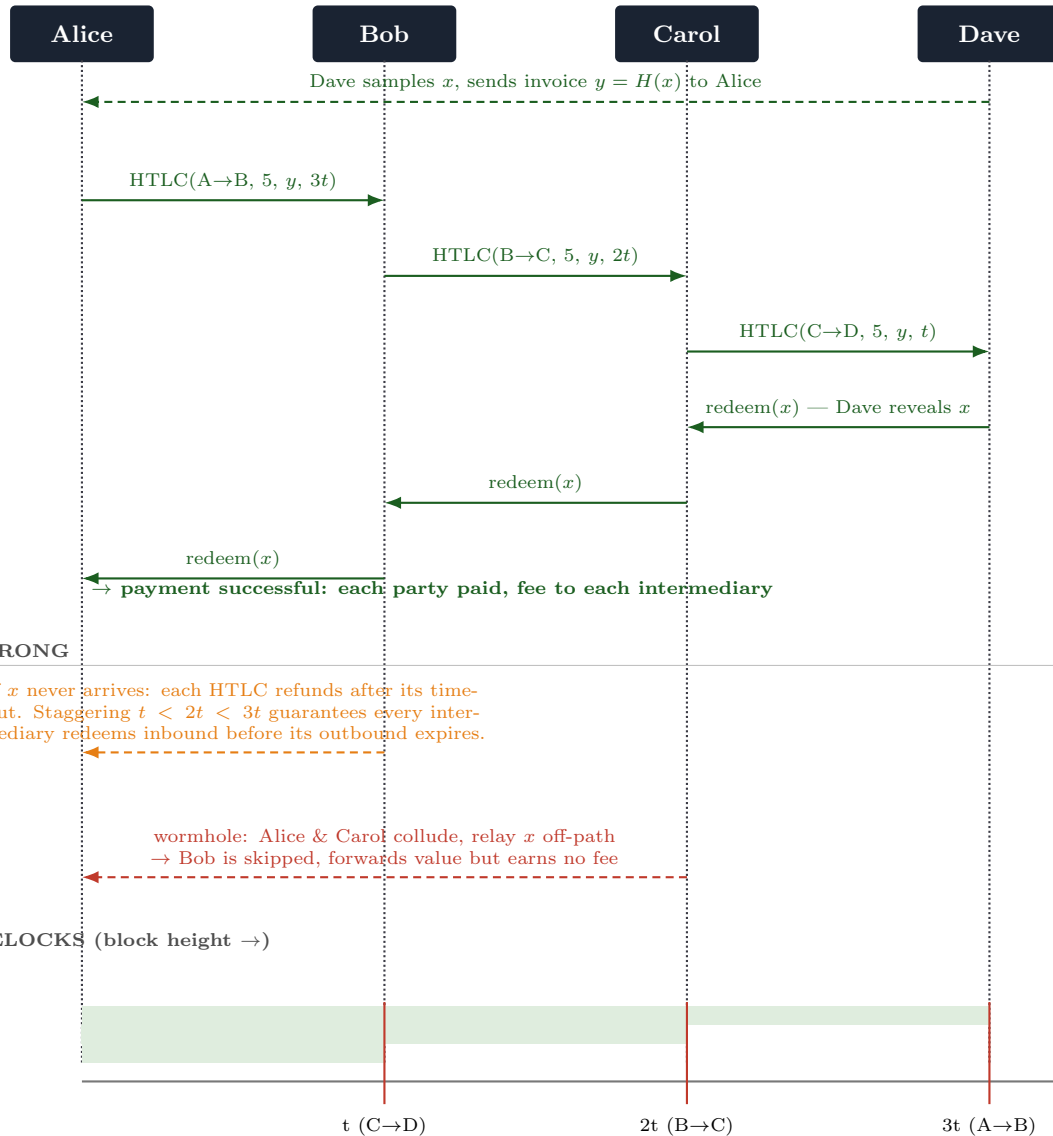


Multi-hop payment: Alice → Bob → Carol → Dave

honest success / safety proved timeout / refund attack (reachable)

annotated with multihop.spthy lemmas — see rail →



LEMMA · multihop.spthy

1 Invoice & preimage authentication

Invoice.Authenticates.Settlement,
Preimage.Secret.Until.Released,
Forged.Invoice.Requires.Key.Compromise,
Invoice.Released.Once ✓

2 Lock causality (forward needs previous)

Forward1.Requires.Offet, Forward2.Requires.Forward1,
HTLC.On.Opened.Channel ✓

3 Release causality (redeem needs preimage)

Fulfill.Requires.Forward2, Claim.Requires.Release,
Settle.Requires.Receiver.Release ✓

4 Atomicity + non-vacuity

Payment.Atomicity.Under.Liveness ✓,
Multihop.Payment.Possible ✓ (witness)

5 Value & fee conservation

EndToEnd.Value.Conservation, Fee.Conservation.Hop1/2,
Fee.Strictly.Positive.Hop1/2,
Receiver.Paid.Invoice.Amount ✓

6 Refund / grieving

Refund.Requires.Timeout, Loss.Requires.Inaction ✓,
Honest.Intermediary.Refunds ✓ (reachable)

7 Wormhole fee theft

Wormhole.Fee.Theft.Reachable ✓,
Wormhole.Steals.Exactly.The.Fees ✓ (theft = path fees)

8 Intermediary safety (under liveness T2b)

Intermediary.Never.Loses.Under.Liveness ✓,
Timeout.Race.Blocked ✓ (race blocked with T2b)

✓ = machine-checked in multihop.spthy (43/43).
green = all-traces safety, orange = benign/reachable,
red = attack shown reachable (by design, no PTLC).

redeem windows(nested)
Channel-layer lemmas (open/update/close/punish)
live on the Alice↔Bob figure.